

LLM Abstract Screening for CONSORT Eligibility: A Solve-or-Guess Validation

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June 25, 2026

1 Introduction

Systematic reviews begin with screening. To decide whether a study belongs in a review, someone reads its title and abstract and judges it against the review’s eligibility criteria. Every systematic review does this, across medicine and beyond, and screening is its most repetitive and labor-intensive stage. The aggregate volume is large: thousands of studies appear in most search strings, and a single review screens far more abstracts than it keeps. Large language models are now used for this step, promising to do in minutes what takes a reviewer hours.

Automating screening is a long-standing goal, and the approaches have evolved. Classical machine learning treated it as semi-automation: an active learner re-ranks the unscreened abstracts as a reviewer labels, surfacing likely includes first so the reviewer can stop early while still catching most of them (Chai et al., 2021). Large language models shifted the design toward full automation. Recent systems use an LLM directly as a zero-shot classifier, and ensembles of several models reach high sensitivity and specificity on large human-curated benchmarks such as the Cochrane reviews (Sanghera et al., 2024). Other work trades binary classification for graded relevance, ranking abstracts against the inclusion criteria with LLM-guided neural re-rankers (Jaumann et al., 2025). These systems differ in design, but they are judged the same way, against a human-curated gold standard.

The metrics are shared too. A screener is scored on sensitivity (recall), specificity, and precision or F1 against the gold standard (Kim et al., 2025; Sanghera et al., 2024). Because a review cannot afford to miss relevant studies, the usual target is near-complete sensitivity, and prompts are tuned to bias the classifier toward inclusion until recall approaches one, accepting many false positives to be cleared later (Sanghera et al., 2024). These

numbers also strain in the setting where screening actually happens. Inclusions are rare and the classes are severely imbalanced, so accuracy is dominated by the easy exclusions and chance-corrected agreement such as Cohen’s κ can read as poor even when raters mostly agree. Recent guidance warns that standard confusion-matrix metrics mislead under exactly these low-prevalence, cost-asymmetric conditions (Madeyski et al., 2025).

A deeper problem is that these frameworks treat the human benchmark as infallible. When a screener and a reviewer disagree, the metrics count it as a screener error, yet they cannot tell an LLM mistake from ordinary disagreement between conscientious reviewers. A single human benchmark cannot separate the two. In practice, validation often leans on a gold standard built by the same reviewers the screener is later compared against, which makes the comparison circular. For an eligibility criterion where experienced reviewers themselves disagree, the circularity is not a corner case. It is the normal situation.

We validate the screener with the solve-or-guess model of Rohe et al. (2026). The model treats the automated screener as one rater among several and asks how often each rater solves the task, rather than how often it matches a designated key. Each rater a has a solving probability p_a . Under the model, Cohen’s κ between two raters factors into the product of their solving probabilities, so the ratio of two raters’ agreement with a common third rater recovers the ratio of their solving probabilities. The comparison needs no gold standard, and it applies to any screening decision that several raters make on the same items. If the screener agrees with each human about as often as the humans agree with one another, it solves the task about as well as they do.

We demonstrate the approach on one screening task with concrete stakes: identifying individually-randomized trials that evaluate patient-level health outcomes, the studies to which the CONSORT reporting checklist applies. CONSORT specifies the minimum information a randomized trial report should contain, such as how randomization was done, who was blinded, and how outcomes were defined. It is endorsed by over 600 journals, and its current version is a 30-item checklist updated in 2025 (Hopewell et al., 2025). Adherence remains poor: authors often certify that they followed CONSORT while the manuscript still omits items that matter for risk-of-bias assessment, which distorts the evidence syntheses that depend on those reports. Tools are emerging to close this gap, including an LLM that drafts CONSORT-completeness review reports inside a journal’s editorial workflow, now under test in a randomized trial at *The BMJ* (Krauska et al., 2026). Each such effort begins by identifying which papers are in scope,

which is the screening task we validate here.

Three experienced reviewers and one LLM labeled the same 100 recent medRxiv abstracts as Include or Exclude for the randomized trial on CONSORT adherence, the LLM working from a codebook prompt at a fixed temperature of 0. We fit the solve-or-guess model to the four raters. The LLM is statistically indistinguishable from the three human reviewers: its estimated solving probability is the highest of the four, and every kappa-ratio interval against a human contains one. This is evidence that, on this task and this corpus, the LLM screens at the level of a human reviewer, established without a gold standard.

This paper provides a repeatable methodology. Solve-or-guess gives a gold-standard-free way to certify an LLM screener against several human reviewers, and CONSORT-eligibility screening is the worked example. The pipeline is reproducible end to end, so another review can rerun the validation on its own eligibility criteria and its own reviewers. Section 2 states the task, the codebook, and the model. Section 3 reports the estimates. Section 4 discusses what the finding does and does not state, for this task and for screening more generally.

2 Methods

We estimate how often an LLM solves the abstract-screening task relative to human reviewers, on the same abstracts and without a gold-standard label set. The estimate comes from the solve-or-guess model (Rohe et al., 2026), which summarizes each rater by a single solving probability and recovers those probabilities from agreement alone. This section states the screening task, recalls the part of the model we use, and gives the estimation procedure. We rely on Rohe et al. (2026) for the model’s development and proofs, and restate only what the reader needs to follow the analysis.

2.1 The screening task

The task is to decide, from a title and abstract, whether a paper belongs to the target population for full-text CONSORT review: individually-randomized trials that evaluate patient-level health outcomes. Each abstract receives one label, Include or Exclude.

2.2 The codebook

The LLM screens each abstract by working through a codebook, the prompt that operationalizes the eligibility criteria. The codebook poses six questions to create chain-of-thought. The first four gather evidence with simple tangible questions: whether the manuscript analyzes empirical data, whether the analysis evaluates patient-level health outcomes, what allocation language the abstract contains, and whether that language indicates individual randomization. The fifth question records the reasoning that synthesizes the evidence. The sixth states the decision: Include, Exclude, or Human Review, the last reserved for abstracts too ambiguous to call. The codebook text is `prompts/screen_v1_codebook.txt`.

We wrote the codebook on a development set, deliberately held apart from the abstracts we evaluate. The development set is the 20 preprints that met the same “random” search criterion and were posted immediately before the evaluation corpus. We drafted and revised the six questions against these 20 abstracts, then froze the codebook before running it on the 100 evaluation abstracts. The two sets are disjoint, so the prompt was never tuned on the abstracts it is later scored against. We hold the codebook fixed for every abstract in the evaluation.

Careful codebook creation is an important implementation step. The comparison rests on solver consensus (Section 2.4): the assumption that each abstract has a single correct label that conscientious raters converge on when they solve. That assumption holds only as long as the codebook is sharp. A vague criterion lets two careful reviewers reach different labels for good reasons, and then there is no single target to recover, so the kappa-ratio no longer measures solving ability. Fixing the eligibility question in six concrete steps is what makes solver consensus plausible here. The codebook is therefore a precondition for the whole comparison, and we treat its design as part of the method.

2.3 The corpus and raters

We drew the evaluation corpus from medRxiv: the 100 most recent preprints whose title or abstract contains the word “random,” retrieved on 1 May 2026. Their posting dates range from 11 April to 30 April 2026.

Four raters labeled the same 100 abstracts: three human reviewers and one LLM. The design is complete. Every rater labeled every abstract, so all $\binom{4}{2} = 6$ rater pairs are observed on all 100 items. The three human reviewers worked from the same codebook as the LLM, though they were not asked

to record the codebook’s chain-of-thought reasoning; each recorded only the final Include or Exclude label. The LLM ran the codebook as a single prompt with Claude Haiku 4.5 at temperature 0, and returned Human Review on 2 of the 100 abstracts. We collapse Human Review to Include, the conservative choice for a screening step that feeds a downstream full-text audit.¹ The label space is then binary for all four raters.

The corpus also postdates the model’s training data, which ends in July 2025; the abstracts were posted in April 2026.² The LLM therefore could not have seen these papers during training, which rules out data contamination: its agreement with the human reviewers cannot be an artifact of having encountered the abstracts before. This is a built-in strength of validating a screener on newly posted preprints.

2.4 The solve-or-guess model

The solve-or-guess model treats each label as the output of one of two latent states (Rohe et al., 2026). Each abstract i has a true label Y_i , here its true CONSORT-eligibility status. On each abstract, each rater a is either solving or guessing. With probability p_a the rater solves and reports the true label, so $R_{a,i} = Y_i$. Otherwise the rater guesses, drawing a label from a rater-specific distribution $\pi^{(a)}$ that does not depend on Y_i . The parameter $p_a \in [0, 1]$ is the rater’s solving probability. It is one number per rater, on the same scale across raters, and it is the quantity we compare.

Two raters can agree because both solved and recovered Y_i , or because at least one guessed and the labels happened to match. The model separates the two. Under the model, Cohen’s κ (Cohen, 1960) between two raters factors into the product of their solving probabilities, $\kappa(a, b) = p_a p_b$ (Rohe et al., 2026). The factorization has a corollary we use directly. For three raters a , b , and c , the ratio $\kappa(a, c)/\kappa(b, c) = p_a/p_b$ recovers the ratio of two raters’ solving probabilities, free of the reference rater c and free of any gold-standard label. We read $p_{\text{LLM}}/p_{\text{human}} > 1$ as the LLM solving the task more often than that human. The ratio is robust to biased guessing: the comparison survives even when the individual κ values are distorted by the raters’ guessing distributions (Rohe et al., 2026). Four raters suffice to identify the solving probabilities, since identification requires three or more (Rohe et al., 2026).

¹With only 2 of 100 abstracts affected, the alternative policies (collapse to Exclude, treat Human Review as a third label class, or drop the item) move the estimates negligibly.

²Claude Haiku 4.5 has a July 2025 training-data cutoff, per Anthropic’s published model documentation.

The model rests on two commitments about our task. First, each abstract admits a single Y_i for solvers to converge on. This is a statement that the codebook is sharp enough that conscientious reviewers would agree on the label. Second, the raters’ labels are independent given Y_i , so that agreement reflects shared recovery of the truth rather than shared bias or copying. The human reviewers labeled independently, and the LLM had no access to their decisions, so the second commitment holds by construction of the study.

2.5 Estimation and inference

We fit the homogeneous model, in which each rater’s solving probability is constant across abstracts. The baseline run carries no per-item difficulty features, so the heterogeneous extension of Rohe et al. (2026) does not apply here. We estimate the parameters $(\tau, \{\pi^{(a)}\}, \{p_a\})$ by maximum likelihood with the EM algorithm of Rohe et al. (2026), treating the LLM as one rater and each human as a separate rater. Each rater receives its own guessing distribution.

We report three quantities. The solving probability $\hat{p}_a$ for each of the four raters orders them on a common scale. The kappa-ratio $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each of the three humans is the primary comparison; an interval that contains 1 means the LLM is indistinguishable from that human. Observed pairwise Cohen’s κ for all six rater pairs is reported as a descriptive check on the raw agreement.

We quantify uncertainty with the nonparametric bootstrap of Rohe et al. (2026), resampling abstracts rather than individual labels so that the agreement structure within each abstract is preserved. We draw $n = 100$ abstracts with replacement, refit the model, and repeat $B = 1000$ times. For each solving probability and each kappa-ratio we report the 95% bootstrap interval.

We check the model’s fit with the rank-one diagnostic of Rohe et al. (2026). Under the model the off-diagonal agreement matrix is rank one plus a chance constant, $A_{ab} = c + (1 - c) p_a p_b$. We measure the departure from that structure by the off-diagonal residual sum of squares $T = \sum_{a < b} (A_{ab} - c - (1 - c) p_a p_b)^2$, and obtain its null distribution by a parametric bootstrap that simulates datasets from the fitted model, refits, and recomputes T ($B = 1000$). A large p-value means the observed agreement is consistent with the model.

The screening run and the analysis are scripted end to end (`code/03_run_screening.py`, `code/04_run_solve_or_guess.R`, and `code/05_rank_one_diagnostic.R`); 04

calls the estimator and the bootstrap from Rohe et al. (2026). Every reported number traces back to the experiment configuration `experiments/exp_001_codebook.yaml`.

3 Results

We fit the homogeneous solve-or-guess model to the four raters on the 100 abstracts. We read off three things: each rater’s solving probability, the kappa-ratio of the LLM against each human, and the observed pairwise agreement. We label the three human reviewers A, B, and C, and write LLM for the automated rater.

3.1 Solving probabilities

Table 1 reports the estimated solving probability for each rater with its 95% bootstrap interval. The LLM has the highest point estimate, 0.85, above all three humans. The four intervals overlap heavily. The LLM’s interval, $[0.69, 1.00]$, contains the point estimate of every human, and each human’s interval contains the LLM’s. On solving probability alone, the LLM sits at the top of the human range rather than outside it.

Table 1: Estimated solving probability $\hat{p}_a$ for each rater, with 95% nonparametric bootstrap intervals ($B = 1000$). The LLM’s estimate sits at the top of the human range.

Rater	$\hat{p}_a$	95% CI
LLM	0.85	$[0.69, 1.00]$
A	0.81	$[0.61, 0.99]$
B	0.83	$[0.64, 0.98]$
C	0.67	$[0.46, 0.88]$

3.2 The LLM against each human

The kappa-ratio compares two raters directly, without a gold standard (Rohe et al., 2026). Table 2 reports $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each reviewer. All three point ratios are at or above 1: 1.05 against A, 1.02 against B, and 1.26 against C. Every interval contains 1. The data do not separate the LLM’s solving probability from any human’s. The widest gap is against reviewer C,

the lowest-scoring human, where the ratio is 1.26, but its interval $[0.88, 1.98]$ still includes 1.

Table 2: Kappa-ratio $\hat{p}_{\text{LLM}}/\hat{p}_{\text{human}}$ against each human reviewer, with 95% bootstrap intervals. An interval containing 1 means the two raters are indistinguishable.

Comparison	Ratio	95% CI
LLM / A	1.05	$[0.81, 1.40]$
LLM / B	1.02	$[0.77, 1.39]$
LLM / C	1.26	$[0.88, 1.98]$

3.3 Observed agreement

Table 3 reports observed pairwise Cohen’s κ for all six rater pairs, a descriptive check on the raw agreement behind the model fit. The values range from 0.61 to 0.79, the moderate-to-substantial band. The LLM’s agreement with the humans (0.61 to 0.79) falls in the same range as the humans’ agreement with one another (0.70 to 0.76). The LLM is not an outlier in the agreement structure.

Table 3: Observed pairwise Cohen’s κ for all six rater pairs.

	A	B	C	LLM
A	—	0.76	0.70	0.79
B		—	0.71	0.73
C			—	0.61
LLM				—

3.4 Model fit

The model assumes the off-diagonal agreement matrix is rank one plus a chance constant. The rank-one diagnostic does not reject this structure (parametric bootstrap $p = 0.38$, $B = 1000$), so the agreement among the four raters is consistent with the solve-or-guess model. With four raters and 100 abstracts the test has limited power, so this is reassurance rather than strong confirmation.

3.5 Summary

The LLM’s solving probability is statistically indistinguishable from each of the three human reviewers. Its point estimate is the highest of the four raters, every kappa-ratio interval against a human contains 1, and its pair-wise agreement with the humans matches the humans’ agreement among themselves. On this screening task, and on this corpus, the LLM screens at the level of a human reviewer.

4 Discussion

This method can be generalized to other validation tasks. Any team comparing an LLM or another machine-learning tool against several human raters on the same items can use solve-or-guess to estimate how well the tool performs, with no gold standard. CONSORT-eligibility screening is the case we worked through. On that task the LLM solves about as often as the human reviewers. This is a model-based reliability claim, following from the solve-or-guess factorization under solver consensus and conditional independence (Rohe et al., 2026), and it is not a claim of gold-standard accuracy. We never measured the LLM against a key. We measured how its agreement with three reviewers compares to their agreement with one another, and the two are indistinguishable.

When an LLM can replace a human rater is a context dependent decision, where risks need to be weighed. In the case of the randomized controlled trial of CONSORT labeling (Krauska et al., 2026), this evidence makes us comfortable doing eligibility screening for the trial with a human reviewer confirming eligibility at the point of enrollment. The numbers are also specific to this task and this corpus. The estimand is tied to one eligibility criterion and to a sample of recent medRxiv preprints, and it should not be carried to a different criterion or a different literature without rerunning the analysis. The method travels; these particular solving probabilities do not.

Two assumptions carry the comparison, and both can fail on a harder task (Rohe et al., 2026). Solver consensus says each abstract has a single correct label that conscientious reviewers converge on, which is a claim that the codebook is sharp; where it is not, the fix is to repair the codebook before grading any rater against it. Conditional independence says agreement comes from shared recovery of the truth, not from shared bias or copying. It holds here by construction, since the LLM never saw the human labels, though a surface cue that both the LLM and the humans key on would still inflate agreement. The rank-one diagnostic of Rohe et al. (2026) tests for

these violations, and it does not reject the model here, though with only four raters it is a weak check.

The study has clear limits. We evaluated a single codebook prompt, with one model, in one pass at temperature 0, so we neither compared prompt variants nor estimated the variance of repeated runs. We fit the homogeneous model, with a constant solving probability per rater, rather than the heterogeneous extension that lets solving probability vary with item difficulty. The corpus is 100 abstracts, so the bootstrap intervals are wide and only sizable differences are detectable. The three reviewers and the LLM worked from the same codebook, which supports solver consensus but also means a systematic gap in the codebook would bias all four raters together and inflate their agreement.

The natural next step is to push the method into settings that strain these assumptions, rather than only tasks as clean as this one. Here the codebook is sharp, the reviewers worked independently, and the LLM never saw their labels. Some reviewer pools train together or confer, and an automated screener may share systematic biases with the humans it is judged against, both of which break conditional independence and inflate agreement. We want to map where the method still gives a defensible answer and where it misleads. Two tools from Rohe et al. (2026) bear on this directly. The rank-one diagnostic tests the assumptions dataset by dataset and flags where the basic model should not be trusted. The multi-community generalization relaxes solver consensus to allow several legitimate labels per item, which is the right model when reviewers disagree for principled reasons. Lighter extensions remain open too: comparing prompt variants by their effect on the kappa-ratio, ordinal screening decisions, and other models and review topics.

Screening sits at the front of every systematic review, and it is a measurement problem with no gold standard. Solve-or-guess lets a review validate an LLM screener against its own disagreeing reviewers and still get a defensible answer. We demonstrated this on CONSORT-eligibility screening, where the LLM screens at the level of a human reviewer and the abstracts that clear the gate feed downstream reporting-quality work. The recipe is not specific to CONSORT. Any review whose reviewers label the same abstracts can certify its screener the same way. As LLMs take on more of the screening load, this is how a review decides whether to trust them.

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